# Modeling Neutron Stars
# Lecture-10

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UNIVERSITÉ CAEN NORMANDIE

**Nuclear Theory Group**
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**Elementary General Theory of Relativity**

- Recap: Equivalence and Geodesics
- General covariance

# Principle of Equivalence

## Matter-space-time

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- **Statement:** *At every spacetime point in an arbitrary gravitational field (anytime anywhere in the universe), a local inertial (Lorentz) frame can be chosen so that the laws of the physics take on the form they have in STR.*

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- So we have *Christoffel Symbols* of first and second kind,

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- In the Newtonian limit, $g_{00} = 1 - 2\frac{GM}{r}$.

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- **Caution!!!!!** The order of derivatives in flat space time does not matter. But it mattes in curved space time.

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- Starting from the derivative of a scalar we can write,

$$\begin{aligned} S_{;\mu;\nu} = S_{;\mu,\nu} - \Gamma^{\alpha}_{\mu\nu} S_{;\alpha} &= S_{,\mu,\nu} - \Gamma^{\alpha}_{\mu\nu} S_{;\alpha} \text{ as } \; S_{;\mu,\nu} = S_{,\mu,\nu} \\ \text{Also} \quad S_{;\mu;\nu} &= S_{;\nu;\mu} \end{aligned}$$

- For vectors and tensors this is not so straight forward!! Let us start from a vector,

$$\begin{aligned} A_{\sigma;\mu} &= A_{\sigma,\mu} - \Gamma^{\lambda}_{\sigma\mu} A_{\lambda} \equiv B_{\sigma\mu} \\ \Rightarrow A_{\sigma;\mu;\nu} &= B_{\sigma\mu;\nu} = B_{\sigma\mu,\nu} - \Gamma^{\alpha}_{\sigma\nu} B_{\alpha\mu} - \Gamma^{\alpha}_{\mu\nu} B_{\alpha\sigma} \\ \Rightarrow A_{\sigma;\mu;\nu} - A_{\sigma;\nu;\mu} &= A_{\rho} R^{\rho}_{\sigma\mu\nu} \end{aligned}$$

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is the *Riemann-Christoffel curvature tensor.*

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is the *Riemann-Christoffel curvature tensor.* It is the only tensor which can be constructed from first and second derivatives of metric tensor.

# Riemann Curvature Tensor

## Symmetries

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  Because of this symmetry, $R^{\mu}{}_{\nu} = R_{\nu}{}^{\mu} = R^{\mu}_{\nu}$.

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  Because of this symmetry, $R^{\mu}{}_{\nu} = R_{\nu}{}^{\mu} = R^{\mu}_{\nu}$. And the *Ricci scalar curvature* is given by,

$$R = g^{\mu\nu} R_{\mu\nu}$$

# Second covariant derivative

*Bianchi identities*

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- For second rank tensor with arbitrary multiplication form of vectors,

$$\left(A_\mu B_\nu\right)_{;\rho;\sigma} = \left(A_{\mu;\rho} B_\nu + A_\mu B_{\nu;\rho}\right)_{;\sigma}$$

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&= A_\mu B_\alpha R^\alpha_{\nu\rho\sigma} + A_\alpha R^\alpha_{\mu\rho\sigma} B_\nu \\
\Rightarrow T_{\mu\nu;\rho;\sigma} - T_{\nu\mu;\rho;\sigma} &= T_{\mu\alpha} R^\alpha_{\nu\rho\sigma} + T_{\alpha\nu} R^\alpha_{\mu\rho\sigma}.
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- Now putting $T_{\mu\nu} = A_{\mu;\nu}$ in the above expression, we get

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- By cyclic permutation of $\nu\rho\sigma$ we add three equations to get

$$LHS = (A_{\mu;\nu;\rho} - A_{\mu;\rho;\nu})_{;\sigma} + (A_{\mu;\sigma;\nu} - A_{\mu;\nu;\sigma})_{;\rho} + (A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho})_{;\nu}$$

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&= A_{\mu;\rho;\sigma} B_\nu + A_\mu B_{\nu;\rho;\sigma} + A_{\mu;\rho} B_{\nu;\sigma} + A_{\mu;\sigma} B_{\nu;\rho} \\
\Rightarrow (A_\mu B_\nu)_{;\rho;\sigma} - (A_\mu B_\nu)_{;\sigma;\rho} &= A_\mu \left(B_{\nu;\rho;\sigma} - B_{\nu;\sigma;\rho}\right) + \left(A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho}\right) B_\nu \\
&= A_\mu B_\alpha R^\alpha_{\nu\rho\sigma} + A_\alpha R^\alpha_{\mu\rho\sigma} B_\nu \\
\Rightarrow T_{\mu\nu;\rho;\sigma} - T_{\nu\mu;\rho;\sigma} &= T_{\mu\alpha} R^\alpha_{\nu\rho\sigma} + T_{\alpha\nu} R^\alpha_{\mu\rho\sigma}.
\end{aligned}$$

- Now putting $T_{\mu\nu} = A_{\mu;\nu}$ in the above expression, we get

$$A_{\mu;\nu;\rho;\sigma} - A_{\nu;\mu;\rho;\sigma} = A_{\mu;\alpha} R^\alpha_{\nu\rho\sigma} + A_{\alpha;\nu} R^\alpha_{\mu\rho\sigma}$$

- By cyclic permutation of $\nu\rho\sigma$ we add three equations to get

$$\begin{aligned}
LHS &= (A_{\mu;\nu;\rho} - A_{\mu;\rho;\nu})_{;\sigma} + (A_{\mu;\sigma;\nu} - A_{\mu;\nu;\sigma})_{;\rho} + (A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho})_{;\nu} \\
&= \left(A_\alpha R^\alpha_{\mu\nu\rho}\right)_{;\sigma} + \left(A_\alpha R^\alpha_{\mu\sigma\nu}\right)_{;\rho} + \left(A_\alpha R^\alpha_{\mu\rho\sigma}\right)_{;\nu} \\
RHS &= A_{\alpha;\nu} R^\alpha_{\mu\rho\sigma} + A_{\alpha;\sigma} R^\alpha_{\mu\nu\rho} + A_{\alpha;\rho} R^\alpha_{\mu\sigma\nu} \text{ using } R^\alpha_{\nu\rho\sigma} + R^\alpha_{\sigma\nu\rho} + R^\alpha_{\rho\sigma\nu} = 0 \\
&\Rightarrow A_\alpha \left(R^\alpha_{\mu\nu\rho;\sigma} + R^\alpha_{\mu\sigma\nu;\rho} + R^\alpha_{\mu\rho\sigma;\nu}\right) = 0
\end{aligned}$$

# Second covariant derivative

*Bianchi identities*

- For second rank tensor with arbitrary multiplication form of vectors,

$$\begin{aligned}
(A_\mu B_\nu)_{;\rho;\sigma} &= (A_{\mu;\rho} B_\nu + A_\mu B_{\nu;\rho})_{;\sigma} \\
&= A_{\mu;\rho;\sigma} B_\nu + A_\mu B_{\nu;\rho;\sigma} + A_{\mu;\rho} B_{\nu;\sigma} + A_{\mu;\sigma} B_{\nu;\rho} \\
\Rightarrow (A_\mu B_\nu)_{;\rho;\sigma} - (A_\mu B_\nu)_{;\sigma;\rho} &= A_\mu \left(B_{\nu;\rho;\sigma} - B_{\nu;\sigma;\rho}\right) + \left(A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho}\right) B_\nu \\
&= A_\mu B_\alpha R^\alpha_{\nu\rho\sigma} + A_\alpha R^\alpha_{\mu\rho\sigma} B_\nu \\
\Rightarrow T_{\mu\nu;\rho;\sigma} - T_{\nu\mu;\rho;\sigma} &= T_{\mu\alpha} R^\alpha_{\nu\rho\sigma} + T_{\alpha\nu} R^\alpha_{\mu\rho\sigma}.
\end{aligned}$$

- Now putting $T_{\mu\nu} = A_{\mu;\nu}$ in the above expression, we get

$$A_{\mu;\nu;\rho;\sigma} - A_{\nu;\mu;\rho;\sigma} = A_{\mu;\alpha} R^\alpha_{\nu\rho\sigma} + A_{\alpha;\nu} R^\alpha_{\mu\rho\sigma}$$

- By cyclic permutation of $\nu\rho\sigma$ we add three equations to get

$$\begin{aligned}
LHS &= (A_{\mu;\nu;\rho} - A_{\mu;\rho;\nu})_{;\sigma} + (A_{\mu;\sigma;\nu} - A_{\mu;\nu;\sigma})_{;\rho} + (A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho})_{;\nu} \\
&= \left(A_\alpha R^\alpha_{\mu\nu\rho}\right)_{;\sigma} + \left(A_\alpha R^\alpha_{\mu\sigma\nu}\right)_{;\rho} + \left(A_\alpha R^\alpha_{\mu\rho\sigma}\right)_{;\nu} \\
RHS &= A_{\alpha;\nu} R^\alpha_{\mu\rho\sigma} + A_{\alpha;\sigma} R^\alpha_{\mu\nu\rho} + A_{\alpha;\rho} R^\alpha_{\mu\sigma\nu} \text{ using } R^\alpha_{\nu\rho\sigma} + R^\alpha_{\sigma\nu\rho} + R^\alpha_{\rho\sigma\nu} = 0 \\
&\Rightarrow A_\alpha \left(R^\alpha_{\mu\nu\rho;\sigma} + R^\alpha_{\mu\sigma\nu;\rho} + R^\alpha_{\mu\rho\sigma;\nu}\right) = 0 = R^\alpha_{\mu\nu\rho;\sigma} + R^\alpha_{\mu\sigma\nu;\rho} + R^\alpha_{\mu\rho\sigma;\nu}
\end{aligned}$$

# Second covariant derivative

*Bianchi identities*

- For second rank tensor with arbitrary multiplication form of vectors,

$$\begin{aligned}
(A_\mu B_\nu)_{;\rho;\sigma} &= (A_{\mu;\rho} B_\nu + A_\mu B_{\nu;\rho})_{;\sigma} \\
&= A_{\mu;\rho;\sigma} B_\nu + A_\mu B_{\nu;\rho;\sigma} + A_{\mu;\rho} B_{\nu;\sigma} + A_{\mu;\sigma} B_{\nu;\rho} \\
\Rightarrow (A_\mu B_\nu)_{;\rho;\sigma} - (A_\mu B_\nu)_{;\sigma;\rho} &= A_\mu \left(B_{\nu;\rho;\sigma} - B_{\nu;\sigma;\rho}\right) + \left(A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho}\right) B_\nu \\
&= A_\mu B_\alpha R^\alpha_{\nu\rho\sigma} + A_\alpha R^\alpha_{\mu\rho\sigma} B_\nu \\
\Rightarrow T_{\mu\nu;\rho;\sigma} - T_{\nu\mu;\rho;\sigma} &= T_{\mu\alpha} R^\alpha_{\nu\rho\sigma} + T_{\alpha\nu} R^\alpha_{\mu\rho\sigma}.
\end{aligned}$$

- Now putting $T_{\mu\nu} = A_{\mu;\nu}$ in the above expression, we get

$$A_{\mu;\nu;\rho;\sigma} - A_{\nu;\mu;\rho;\sigma} = A_{\mu;\alpha} R^\alpha_{\nu\rho\sigma} + A_{\alpha;\nu} R^\alpha_{\mu\rho\sigma}$$

- By cyclic permutation of $\nu\rho\sigma$ we add three equations to get

$$\begin{aligned}
LHS &= (A_{\mu;\nu;\rho} - A_{\mu;\rho;\nu})_{;\sigma} + (A_{\mu;\sigma;\nu} - A_{\mu;\nu;\sigma})_{;\rho} + (A_{\mu;\rho;\sigma} - A_{\mu;\sigma;\rho})_{;\nu} \\
&= \left(A_\alpha R^\alpha_{\mu\nu\rho}\right)_{;\sigma} + \left(A_\alpha R^\alpha_{\mu\sigma\nu}\right)_{;\rho} + \left(A_\alpha R^\alpha_{\mu\rho\sigma}\right)_{;\nu} \\
RHS &= A_{\alpha;\nu} R^\alpha_{\mu\rho\sigma} + A_{\alpha;\sigma} R^\alpha_{\mu\nu\rho} + A_{\alpha;\rho} R^\alpha_{\mu\sigma\nu} \text{ using } R^\alpha_{\nu\rho\sigma} + R^\alpha_{\sigma\nu\rho} + R^\alpha_{\rho\sigma\nu} = 0 \\
&\Rightarrow A_\alpha \left(R^\alpha_{\mu\nu\rho;\sigma} + R^\alpha_{\mu\sigma\nu;\rho} + R^\alpha_{\mu\rho\sigma;\nu}\right) = 0 = R^\alpha_{\mu\nu\rho;\sigma} + R^\alpha_{\mu\sigma\nu;\rho} + R^\alpha_{\mu\rho\sigma;\nu}
\end{aligned}$$

These are called *Bianchi identities*.

# Einstein Tensor

*Vanishing divergence!*

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$0 = g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right)$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$
\begin{aligned}
0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\
&= \left(g^{\mu\nu} R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu} R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu}
\end{aligned}
$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu} R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu} R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu} R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu} g^{\alpha\beta} R_{\beta\mu\nu\rho}$$

# Einstein Tensor

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- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu}R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\beta\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\mu\beta\rho\nu}$$

# Einstein Tensor

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- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu} R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu} R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu} R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu} g^{\alpha\beta} R_{\beta\mu\nu\rho} = g^{\mu\nu} g^{\alpha\beta} R_{\mu\beta\rho\nu} = g^{\alpha\beta} R^{\nu}_{\beta\rho\nu}$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu}R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\beta\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\mu\beta\rho\nu} = g^{\alpha\beta}R^{\nu}_{\beta\rho\nu} = g^{\alpha\beta}R_{\beta\rho}$$

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$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu}R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\beta\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\mu\beta\rho\nu} = g^{\alpha\beta}R^{\nu}_{\beta\rho\nu} = g^{\alpha\beta}R_{\beta\rho} = R^{\alpha}_{\rho}$$

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$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu}R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\beta\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\mu\beta\rho\nu} = g^{\alpha\beta}R^{\nu}_{\beta\rho\nu} = g^{\alpha\beta}R_{\beta\rho} = R^{\alpha}_{\rho}$$

$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha}$$

# Einstein Tensor

*Vanishing divergence!*

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$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

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$$g^{\mu\nu}R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\beta\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\mu\beta\rho\nu} = g^{\alpha\beta}R^{\nu}_{\beta\rho\nu} = g^{\alpha\beta}R_{\beta\rho} = R^{\alpha}_{\rho}$$

$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu}R_{\mu\nu}$$

# Einstein Tensor

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$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

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$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu}R_{\mu\nu} = -R$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu} R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu} R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu} R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu} g^{\alpha\beta} R_{\beta\mu\nu\rho} = g^{\mu\nu} g^{\alpha\beta} R_{\mu\beta\rho\nu} = g^{\alpha\beta} R^{\nu}_{\beta\rho\nu} = g^{\alpha\beta} R_{\beta\rho} = R^{\alpha}_{\rho}$$

$$g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu} R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu} R_{\mu\nu} = -R$$

$$g^{\mu\nu} R^{\alpha}_{\mu\rho\alpha} = g^{\mu\nu} R_{\mu\rho}$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu} R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu} R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

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$$g^{\mu\nu} R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu} R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu} R_{\mu\nu} = -R$$

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# Einstein Tensor

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$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

- Now the terms inside the brackets are evaluated as

$$g^{\mu\nu}R^{\alpha}_{\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\beta\mu\nu\rho} = g^{\mu\nu}g^{\alpha\beta}R_{\mu\beta\rho\nu} = g^{\alpha\beta}R^{\nu}_{\beta\rho\nu} = g^{\alpha\beta}R_{\beta\rho} = R^{\alpha}_{\rho}$$
$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu}R_{\mu\nu} = -R$$
$$g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha} = g^{\mu\nu}R_{\mu\rho} = R^{\nu}_{\rho}$$

- Combining these three together one gets

$$R^{\alpha}_{\rho;\alpha} - R_{;\rho} + R^{\nu}_{\rho;\nu}$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

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$$g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha} = g^{\mu\nu}R_{\mu\rho} = R^{\nu}_{\rho}$$

- Combining these three together one gets

$$R^{\alpha}_{\rho;\alpha} - R_{;\rho} + R^{\nu}_{\rho;\nu} = 2R^{\alpha}_{\rho;\alpha} - R_{;\rho} = 0$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

$$\begin{aligned} 0 &= g^{\mu\nu}\left(R^{\alpha}_{\mu\nu\rho;\sigma} + R^{\alpha}_{\mu\sigma\nu;\rho} + R^{\alpha}_{\mu\rho\sigma;\nu}\right) \\ &= \left(g^{\mu\nu}R^{\alpha}_{\mu\nu\rho}\right)_{;\alpha} + \left(g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu}\right)_{;\rho} + \left(g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha}\right)_{;\nu} \end{aligned}$$

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$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu}R_{\mu\nu} = -R$$

$$g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha} = g^{\mu\nu}R_{\mu\rho} = R^{\nu}_{\rho}$$

- Combining these three together one gets

$$R^{\alpha}_{\rho;\alpha} - R_{;\rho} + R^{\nu}_{\rho;\nu} = 2R^{\alpha}_{\rho;\alpha} - R_{;\rho} = 0$$

$$g^{\mu\rho}R^{\alpha}_{\rho;\alpha} = \left(g^{\mu\rho}R^{\alpha}_{\rho}\right)_{;\alpha}$$

# Einstein Tensor

*Vanishing divergence!*

- Starting from the *Bianchi identities*, contracting with the metric tensor

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$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu}R_{\mu\nu} = -R$$
$$g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha} = g^{\mu\nu}R_{\mu\rho} = R^{\nu}_{\rho}$$

- Combining these three together one gets

$$R^{\alpha}_{\rho;\alpha} - R_{;\rho} + R^{\nu}_{\rho;\nu} = 2R^{\alpha}_{\rho;\alpha} - R_{;\rho} = 0$$
$$g^{\mu\rho}R^{\alpha}_{\rho;\alpha} = \left(g^{\mu\rho}R^{\alpha}_{\rho}\right)_{;\alpha} = R^{\mu\alpha}_{;\alpha}$$

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$$g^{\mu\nu}R^{\alpha}_{\mu\rho\alpha} = g^{\mu\nu}R_{\mu\rho} = R^{\nu}_{\rho}$$

- Combining these three together one gets

$$R^{\alpha}_{\rho;\alpha} - R_{;\rho} + R^{\nu}_{\rho;\nu} = 2R^{\alpha}_{\rho;\alpha} - R_{;\rho} = 0$$
$$g^{\mu\rho}R^{\alpha}_{\rho;\alpha} = \left(g^{\mu\rho}R^{\alpha}_{\rho}\right)_{;\alpha} = R^{\mu\alpha}_{;\alpha} = R^{\mu\nu}_{;\nu}$$

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$$g^{\mu\nu}R^{\alpha}_{\mu\alpha\nu} = -g^{\mu\nu}R^{\alpha}_{\mu\nu\alpha} = -g^{\mu\nu}R_{\mu\nu} = -R$$
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This $G^{\mu\nu}$ is called *Einstein curvature tensor*.

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- The non-zero *Ricci tensor* elements are written finally as $(a' \equiv \frac{\partial a}{\partial r})$,

$$R_{00} = \left(-a'' + b'a' - a'^2 - \frac{2a'}{r}\right)e^{2(a-b)},\quad R_{11} = a'' - b'a' + a'^2 - \frac{2b'}{r},$$
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- *Schwarzchild metric* has singularity at $r = r_S = 2GM$. If this lies outside a star, it is a black hole!!

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## Inside star

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- Eventually one gets

$$\frac{d\mathtt{p}}{dr} = -\frac{[\mathtt{p}(r) + \epsilon(r)][M(r) + 4\pi r^3 \mathtt{p}(r)]}{r[r - 2M(r)]}$$

This is *Tolman-Oppenheimer-Volkoff* equation ($G = c = 1$)

# References:


- Compact Stars, Springer, **Norman K. Glendenning**.
- Quantum Field Theory, Cambridge University Press, **Lewis H. Ryder**.